

Jesse Shen

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Education

Carnegie Mellon University

Master of Science in Biomedical Engineering

Expected: August 2023

GPA: 3.76/4.00

• **Relevant Coursework:** Immunology, Medical Device Innovation and Realization, Introduction to Machine Learning, Biomaterial Host Interactions in Regenerative Medicine, Advanced Physiology

Washington University in St. Louis

Bachelor of Science in Biomedical Engineering

May 2020

GPA: 3.65/4.00

• **Relevant Coursework:** Biomaterials Science, Biomacromolecule Engineering, Molecular and Cellular Engineering, Organic Chemistry

Experience

University of Minnesota- Department of Mechanical Engineering

Research Technician

October 2020 -May 2022

Minneapolis, MN

Thermal contrast amplification for use in lateral flow assays / VigilantDx Startup

- **fM-aM Detection of the SARS-CoV-2 Antigen by Advanced Lateral Flow Immunoassay Based on Gold Nanospheres.**
Yilin Liu, Li Zhan, Jesse W. Shen, Bárbara Baro, Andrea Alemany, James Sackrison, Oriol Mitjà, and John C. Bischof. *ACS Applied Nano Materials* 2021 4 (12), 13826-13837 DOI:[10.1021/acsanm.1c03217](https://doi.org/10.1021/acsanm.1c03217)
- Developed assay to verify higher antibody effectiveness between conjugation methods using a cleavable fluorophore-antigen conjugate.
- Performed antibody conjugation for use in lab-based lateral flow assays involving UV-VIS, DLS, and PAGE.
- Reduced computation times by 50% in Matlab by modifying functions.
- Assembled lateral flow assays with lateral flow reagent dispenser.
- Served as lead author in VigilantDx LFA reader v1.0.16 user manual and produced promotional material.

Organ cryopreservation

- Assisted in the preparation of critical reagents for use in cryopreserving organs such as hearts, kidneys, and livers. These reagents included various cocktails to form highly osmotic, cryoprotective agents that protected organs from ice formation and silica-coated iron oxide nanoparticles that allowed for uniform heating of organs from vitrification.

Academic Research Experience

Carnegie Mellon University Department of Biomedical Engineering

Graduate Student Researcher

July 2022 - January 2023

Pittsburgh, PA

Thymus-on-a-chip

- Developed PDMS organ-on-a-chip design as part of an in vitro microfluidic thymus device in a collaborative project with immunologists.
- Ran flow cytometry to characterize clonogenic thymic epithelial cells adhered to feeder cells

Extracellular Vesicle (EV)-based Cancer Diagnostics

- Cultured non-adherent SY5Y neuroblastoma cells and ran qPCR and ddPCR to lead integration of next-generation sequencing into diagnostic workflow.

Academic projects

Carnegie Mellon University: *Medical Device Innovation and Realization*

Group Technical Lead

January 2023 - May 2023

Pittsburgh, PA

Cell encapsulation for Type-1 diabetes therapy

- Used Solidworks to design a 3D printed model of our cell encapsulation device involving Cura and FDM

Technical Skills

Academic Research: immunoassays · flow cytometry · PCR · mammalian cell culture · microfluidics · protein characterization · PAGE · machine learning · device development · SOP writing · nanoparticle synthesis (ultracentrifugation and vacuum distillation) and conjugation · cryopreservation · 3D printing · CAD

Programming: Matlab · Python